

OBED ARMANDO CAMPOS

FLUID SYSTEMS ENGINEER | EXPERIMENTAL FLUID MECHANICS | TEST & DEVELOPMENT

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PROFESSIONAL SUMMARY

Mechanical engineer with a Ph.D. in Engineering Sciences (Mechanical Engineering) with 5+ years of experience designing, fabricating, modeling, and validating fluid systems and experimental hardware. Background includes pulsatile and oscillatory internal flows, pressure-drop and wall-shear analysis, fluid-structure interaction, CFD, flow diagnostics, and custom test-rig development. Proven ability to translate research requirements into manufacturable hardware, execute verification tests, troubleshoot system performance, and communicate results across multidisciplinary teams.

CORE ENGINEERING EXPERTISE

Fluid-system design & test • Unsteady/internal-flow analysis • Pressure loss & wall shear • Flow-loop and test-fixture development • CFD and analytical modeling • Verification & validation • Experimental uncertainty • Root-cause troubleshooting • MRI and particle-tracking flow diagnostics • CAD, machining, assembly, and laboratory safety

FLUID SYSTEMS & ENGINEERING EXPERIENCE

Graduate Student Researcher | Multiscale Flow Physics Laboratory, UCSD

UC San Diego, CA

Apr 2022 - Jun 2026

- Designed, fabricated, and commissioned a canonical spinal-canal flow phantom and oscillatory flow system to reproduce cerebrospinal-fluid motion for MRI-based flow measurement.
- Integrated experimental measurements with analytical and computational models to characterize unsteady internal flows and evaluate cycle-to-cycle transport behavior.
- Developed MATLAB and Python image-processing workflows for spatial calibration, displacement extraction, data reduction, and comparison across operating conditions.
- Quantified uncertainty and diagnosed limitations associated with signal-to-noise ratio, image resolution, and post-processing, translating findings into test constraints and design recommendations.

Graduate Student Researcher | Experimental Bio-Fluidics Laboratory, UCSD

UC San Diego, CA

Sep 2021 - Jun 2026

- Led the design, build, and test of a multi-chamber pulsatile-flow system that creates controlled pressure and wall-shear environments across multiple test sections for parallel experiments.
- Created analytical and ANSYS Fluent models to predict velocity profiles, pressure losses, and wall shear; used experimental results to verify model trends and refine geometry and operating conditions.
- Developed a 100-mm-wide flow chamber with a stretching membrane to study coupled pressure-driven and oscillatory flow; derived unsteady-flow models and validated performance using dye and particle-tracking measurements.
- Owned end-to-end hardware development, including requirements definition, CAD, fabrication planning, machining coordination, assembly, leak/debug testing, experimental procedures, and data analysis.
- Served as Area Safety Coordinator and mentored mechanical-engineering capstone teams on design reviews, test planning, fabrication, and troubleshooting.

Researcher / Engineering Consultant | Lasheras-del Álamo Biomechanics Laboratory, UCSD

UC San Diego, CA

Feb 2019 - Jun 2021

- Designed and built custom biomechanics test fixtures and flow/imaging devices, converting investigator requirements into manufacturable CAD and experimental hardware.
- Supported fabrication, assembly, troubleshooting, and iterative design improvements for leukocyte-transmigration imaging and related biomechanical studies.
- Advised graduate researchers on experimental-device development, hardware integration, and test-method selection.

EDUCATION

Ph.D., Engineering Sciences (Mechanical Engineering) | University of California, San Diego

San Diego, CA

Jun 2026

- Research focus: fluid-system design, experimental fluid mechanics, pulsatile and oscillatory flows, biofluid mechanics, and multiphysics validation.

M.S., Engineering Sciences (Mechanical Engineering) | UC San Diego | Dec 2023

B.S., Mechanical Engineering - Controls and Robotics Specialization | UC San Diego | Sep 2020

SELECTED PUBLICATIONS & TECHNICAL COMMUNICATION

- Campos, O. et al. "Multi-Chamber Pulsating Flow Device." Journal of Biomechanical Engineering, 2025. DOI: 10.1115/1.4066800.
- Campos, O. et al. "Phantom-Based Evaluation of Time-SLIP MRI for Measuring Spinal-Canal Lagrangian Drift." Submitted to Scientific Reports, 2026.
- Sincomb, S. et al. "Oscillatory Flow in the Cerebral Aqueduct." European Journal of Mechanics - B/Fluids, 2024.
- Schwartz, A. B. et al. "Leukocyte Transmigration Imaging." Frontiers in Cell and Developmental Biology, 2021.
- Presented fluid-mechanics research at the APS Division of Fluid Dynamics Annual Meeting in 2022, 2023, and 2024.

TECHNICAL SKILLS

Fluid Engineering: Internal and unsteady flow; pulsatile and oscillatory systems; pressure-drop and wall-shear calculations; fluid-structure interaction; flow-loop design; test planning; verification and validation; uncertainty analysis.

Modeling & Data: ANSYS Fluent, ANSYS Structural, MATLAB, Python, particle-tracking velocimetry, MRI flow imaging, image processing, experimental data reduction.

CAD & Fabrication: SolidWorks, Inventor, AutoCAD, Fusion 360; CNC, milling, lathe, laser cutting, plastic molding, drill press, system assembly and troubleshooting.

Programming & Controls: MATLAB, Python, C++, Arduino, Linux.

LEADERSHIP & ADDITIONAL EXPERIENCE

- Area Safety Coordinator, Experimental Bio-Fluidics Laboratory, 2023 - 2025.
- Mentor, UCSD Mechanical Engineering Capstone Teams, GEAR, and ISRP programs, 2023 - 2025.
- Teaching Assistant / Reader, UC San Diego, 2021 - 2025; communicated engineering analysis and guided students through technical problem solving.
- Executive Board Member, Latine/x Scholar Network, 2024 - Present.

LANGUAGES

English (native); Spanish (California State Seal of Biliteracy).